

FairLens — Bias Mitigation Report

📅 07 Jul 2026 🎯 Target: **income** 🛡️ Protected: **race** ⭐ Privileged: **White** 📊 Rows: **48,842**

EXECUTIVE SUMMARY

-0.1014 Baseline SPD	-0.0354 Final SPD (RF)	65.1% Bias Reduction	0.8389 Best Accuracy
--------------------------------	----------------------------------	--------------------------------	--------------------------------

FULL PIPELINE RESULTS

Model	Stage	Accuracy	SPD	SPD Status	DI	DI Status	AOD	EOD
RF	Baseline	0.8566	-0.0930	✗ Unfair	0.5651	✗ Unfair	-0.0413	-0.0489
	Reweighting	0.8558	-0.0893	✗ Unfair	0.5820	✗ Unfair	-0.0341	-0.0365
	Reweigh+ADB	0.8512	-0.0633	✗ Unfair	0.6999	✗ Unfair	0.0004	0.0142
	★ Full Pipeline	0.8389	-0.0354	✓ Fair	0.7958	✗ Unfair	0.0457	0.0955
XGBoost	Baseline	0.8713	-0.0993	✗ Unfair	0.5468	✗ Unfair	-0.0486	-0.0650
	Reweighting	0.8708	-0.0586	✗ Unfair	0.7211	✗ Unfair	0.0217	0.0495
	Reweigh+ADB	0.8512	-0.0633	✗ Unfair	0.6999	✗ Unfair	0.0004	0.0142
	★ Full Pipeline	0.8389	-0.0354	✓ Fair	0.7958	✗ Unfair	0.0457	0.0955
TabNet	Baseline	0.8493	-0.0864	✗ Unfair	0.5363	✗ Unfair	-0.0539	-0.0812
	Reweighting	0.8456	-0.0389	✓ Fair	0.7760	✗ Unfair	0.0319	0.0634
	Reweigh+ADB	0.8512	-0.0633	✗ Unfair	0.6999	✗ Unfair	0.0004	0.0142
	★ Full Pipeline	0.8389	-0.0354	✓ Fair	0.7958	✗ Unfair	0.0457	0.0955
LightGBM	Baseline	0.8733	-0.0946	✗ Unfair	0.5568	✗ Unfair	-0.0441	-0.0607
	Reweighting	0.8716	-0.0534	✗ Unfair	0.7428	✗ Unfair	0.0259	0.0523
	Reweigh+ADB	0.8512	-0.0633	✗ Unfair	0.6999	✗ Unfair	0.0004	0.0142
	★ Full Pipeline	0.8389	-0.0354	✓ Fair	0.7958	✗ Unfair	0.0457	0.0955

Note: *Reweight+ADB* uses AIF360 Adversarial Debiasing, which trains a standalone classifier independent of the base model, and CEO is calibrated on its output. So the *Reweight+ADB* and *Full Pipeline (CEO)* rows are identical across all models by design — the base model only affects the Baseline and Reweighting stages.

SMOTE OVERSAMPLING RESULTS

Model	Variant	Accuracy	SPD	DI	AOD	EOD
RF	No SMOTE	0.8566	-0.0930	0.5651	-0.0413	-0.0489
	ADASYN	0.8498	-0.1158	0.5104	-0.0669	-0.0825
	Standard	0.8526	-0.1099	0.5300	-0.0569	-0.0675
	Borderline	0.8509	-0.1143	0.5165	-0.0669	-0.0853
	KMeans	0.8526	-0.1099	0.5300	-0.0569	-0.0675
XGBoost	No SMOTE	0.8713	-0.0993	0.5468	-0.0486	-0.0650
	ADASYN	0.8716	-0.1134	0.5106	-0.0712	-0.1051
	Standard	0.8720	-0.1098	0.5167	-0.0630	-0.0897
	Borderline	0.8705	-0.1088	0.5249	-0.0545	-0.0699
	KMeans	0.8720	-0.1098	0.5167	-0.0630	-0.0897
TabNet	No SMOTE	0.8493	-0.0864	0.5363	-0.0539	-0.0812
	ADASYN	0.8281	-0.1470	0.5102	-0.1003	-0.1229
	Standard	0.8432	-0.1208	0.5409	-0.0709	-0.0882
	Borderline	0.6487	-0.0412	0.9184	-0.0110	-0.0308
	KMeans	0.8432	-0.1208	0.5409	-0.0709	-0.0882
LightGBM	No SMOTE	0.8733	-0.0946	0.5568	-0.0441	-0.0607
	ADASYN	0.8714	-0.1226	0.4673	-0.0826	-0.1189
	Standard	0.8717	-0.1159	0.4921	-0.0693	-0.0960
	Borderline	0.8710	-0.1199	0.4783	-0.0764	-0.1075
	KMeans	0.8717	-0.1159	0.4921	-0.0693	-0.0960

FEATURE IMPORTANCE (TOP 10)

Feature	Importance	Visual
fnlwgt	17.55%	
age	15.60%	
capital_gain	12.06%	
relationship	9.91%	
education_num	8.63%	
hours_per_week	8.42%	
marital_status	6.83%	
occupation	6.57%	
workclass	3.78%	

capital_loss	3.77%
--------------	-------

CONFIGURATION

Parameter	Value
Models	RF, XGBoost, TabNet, LightGBM
Mitigation Steps	Reweighting → ADB → CEO
SMOTE Variants	ADASYN, Standard, Borderline, KMeans
Test Split	30%
Feature Count	13
Total Rows	48,842
Class Distribution	0: 37,155 · 1: 11,687

Generated by FairLens · Fairness detection and mitigation framework · Fair zones: |SPD| < 0.05 · DI ∈ [0.8, 1.25] · |AOD| < 0.1 · |EOD| < 0.1